

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 11

Cardiovascular System

Chapter Goals (Slide 1 of 2)

- Name the parts of the heart and associated blood vessels and their functions in the circulation of blood.
- Trace the pathway of blood through the heart.
- Identify and describe major pathologic conditions affecting the heart and blood vessels.

Chapter Goals (Slide 2 of 2)

- Define combining forms that relate to the cardiovascular system.
- Describe important laboratory tests and clinical procedures pertaining to the cardiovascular system and recognize relevant abbreviations.
- Apply your new knowledge to understand medical terms in their proper context, such as in medical reports and records.

Chapter 11

Lesson 11.1

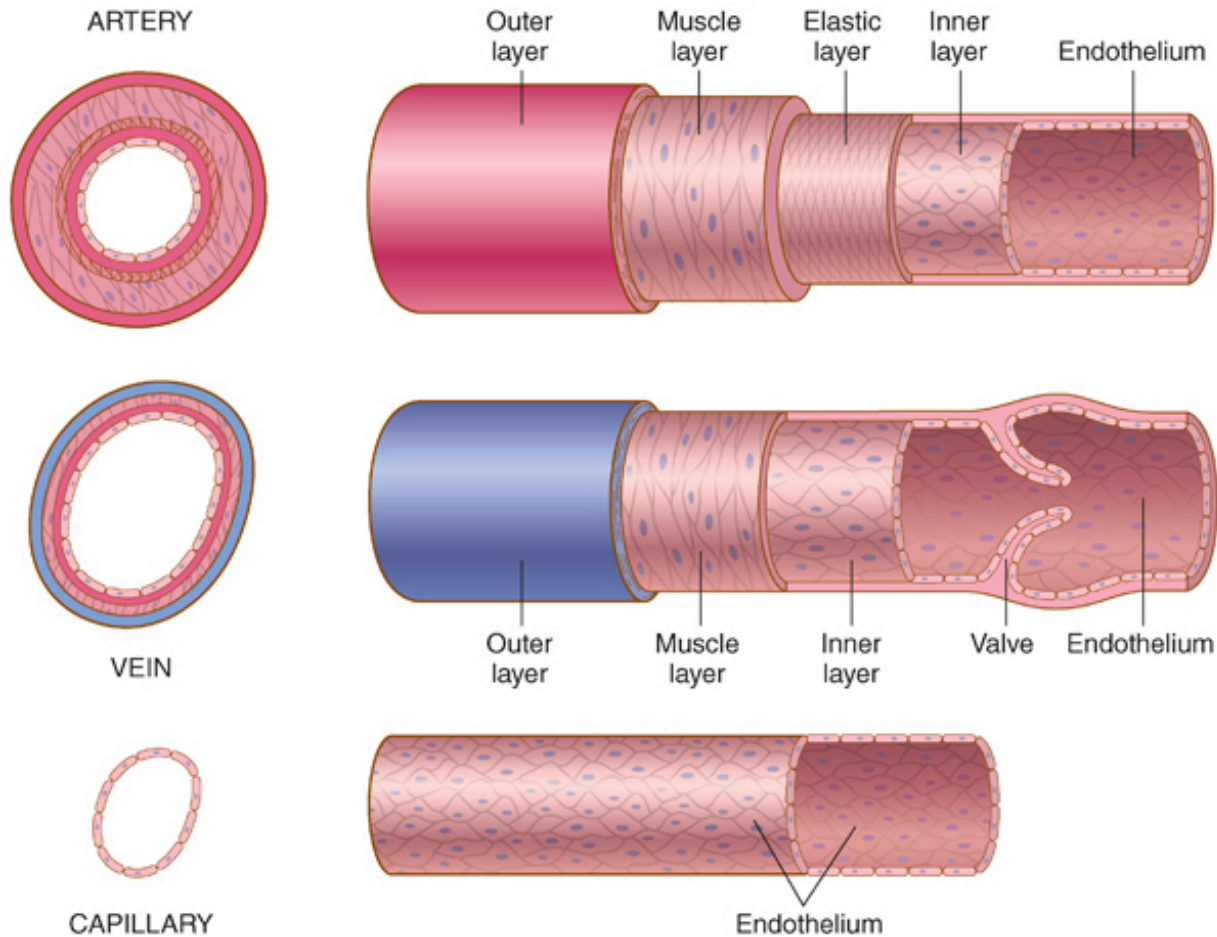
Introduction

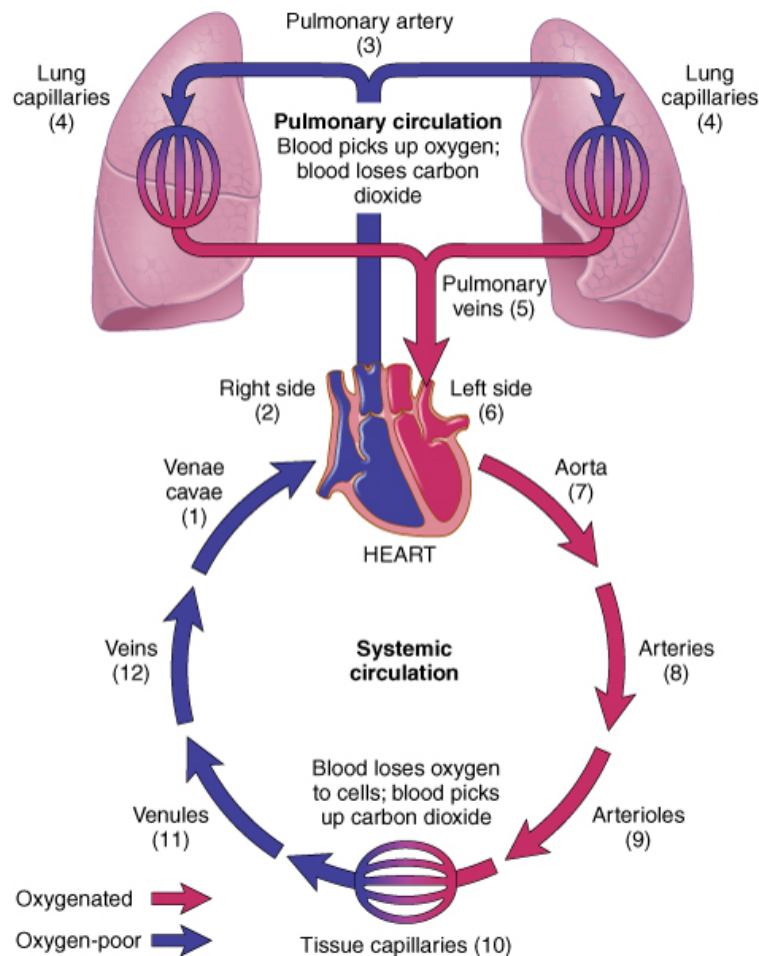
- **Cardiovascular system:** delivers oxygen and nutrients to cells of body tissue
 - Heart (muscular pump)
 - Blood vessels (fuel line and transportation network)

Blood Vessels and the Circulation of Blood

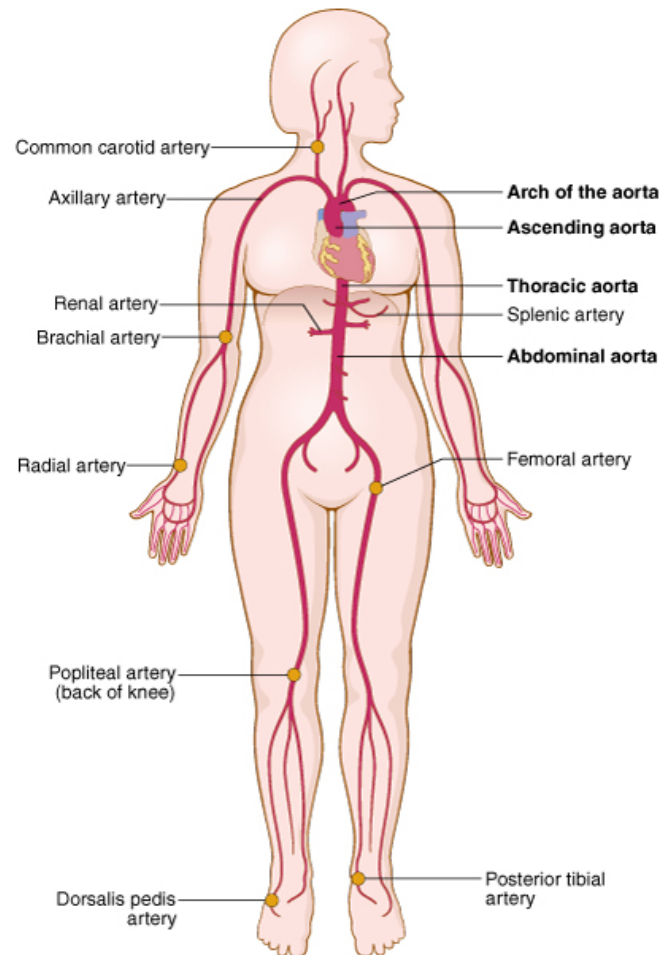
- **Arteries** are the vessels that lead away from the heart.
- **Veins** have thinner walls than arteries and move deoxygenated blood toward the heart from the tissues.
- **Capillaries** are the smallest vessels. They form the point of exchange for oxygen and nutrients into body cells and waste products coming from body cells.

Blood Vessels





The Aorta and Arteries



Anatomy of the Heart

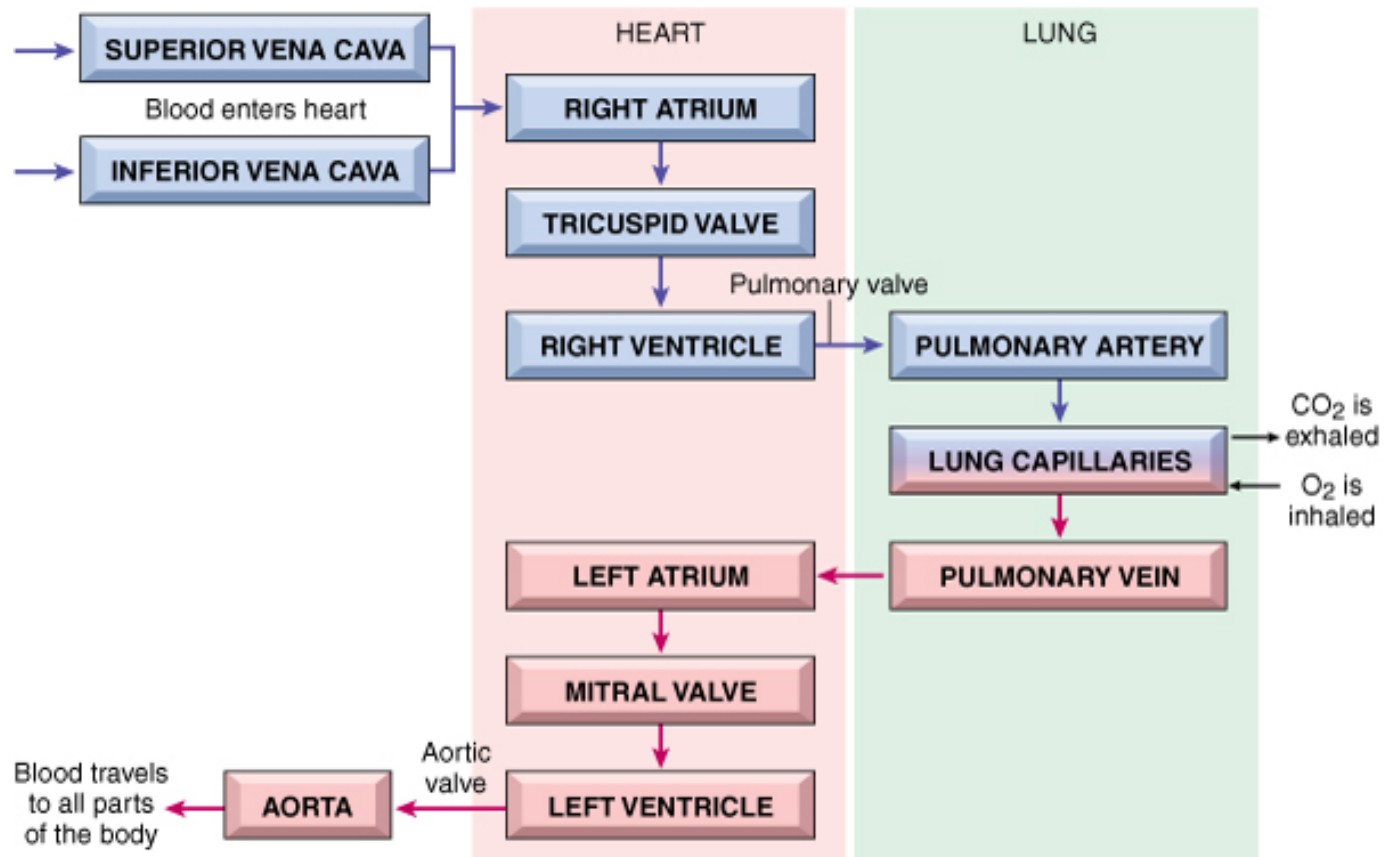
Anatomical structures of the heart include:

- Superior vena cava
- Inferior vena cava
- Right atrium
- Tricuspid valve
- Right ventricle
- Pulmonary valve
- Pulmonary artery
- Pulmonary veins
- Left atrium
- Mitral valve
- Left ventricle
- Aortic valve
- Aorta

Major Valves of the Heart

- **Tricuspid valve** (cusps are flaps of the valves): between the right atrium and right ventricle
- **Pulmonary valve:** between the right ventricle and pulmonary artery
- **Mitral valve:** between the left atrium and left ventricle
- **Aortic valve:** between the left ventricle and aorta

Pathway of Blood through the Heart



Heartbeat

Two phases of the heartbeat:

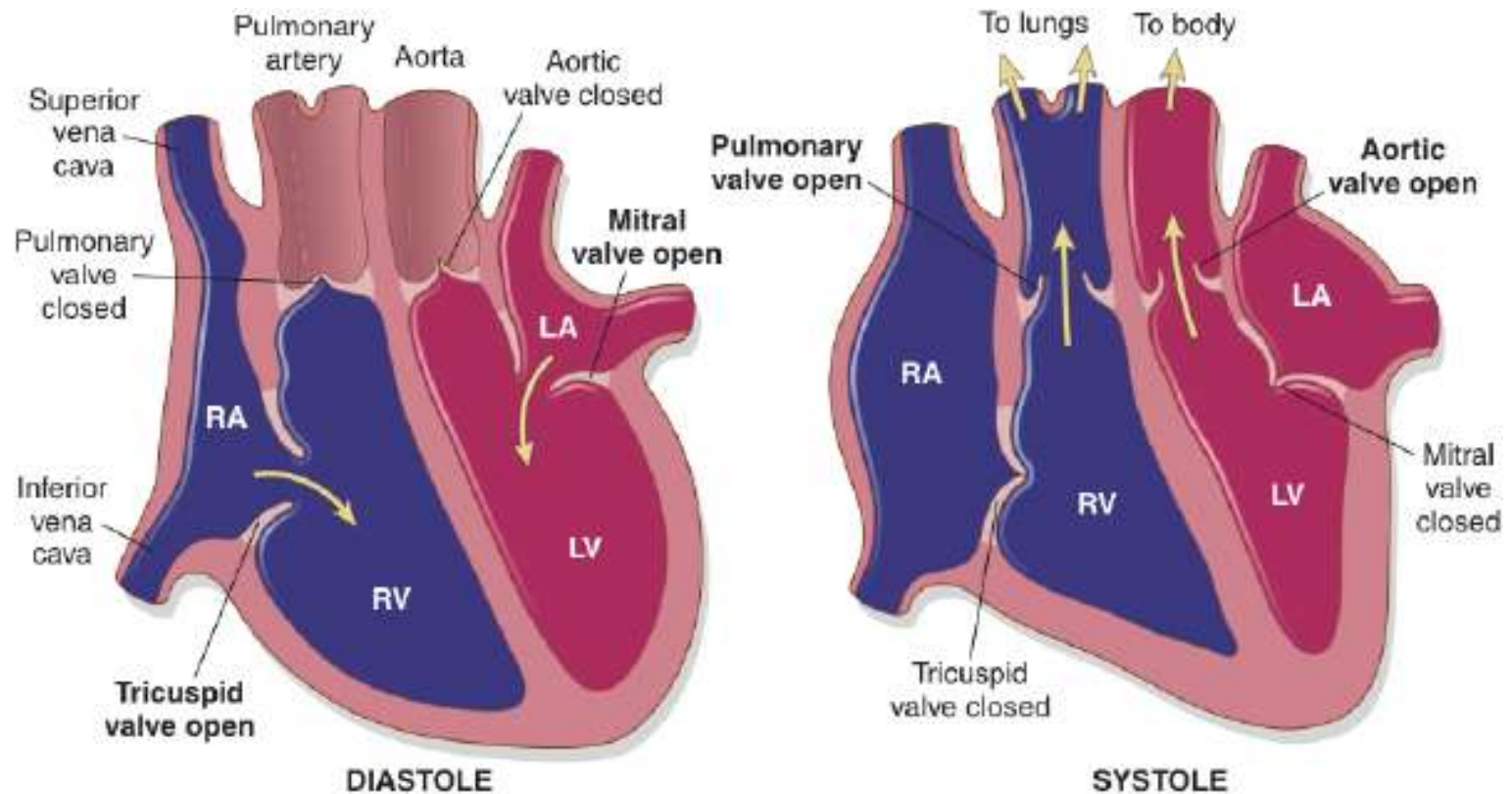
- **Diastole:** relaxation
- **Systole:** contraction
 - The diastole-systole cardiac cycle occurs between 70 to 80 times per minute (100,000 times per day).
 - The heart pumps 3 ounces of blood with each contraction. This means that about 5 quarts are pumped per minute (75 gallons an hour and about 2000 gallons a day).

Heart Sounds

Closure of valves associated with sounds “lubb-dubb, lubb-dubb”

- **lubb**: closure of the tricuspid and mitral valves at the beginning of systole
- **dubb**: closure of the aortic and pulmonary valves at the end of systole
- **murmur**: an abnormal heart sound caused by improper valve closure

Phases of the Heartbeat



Conduction System of the Heart

(slide 1 of 2)

Sinoatrial node (SA node): the pacemaker of the heart

Pacemaker: origin of electrical impulse causing walls of the atria to contract and force blood into the ventricles (ending diastole)

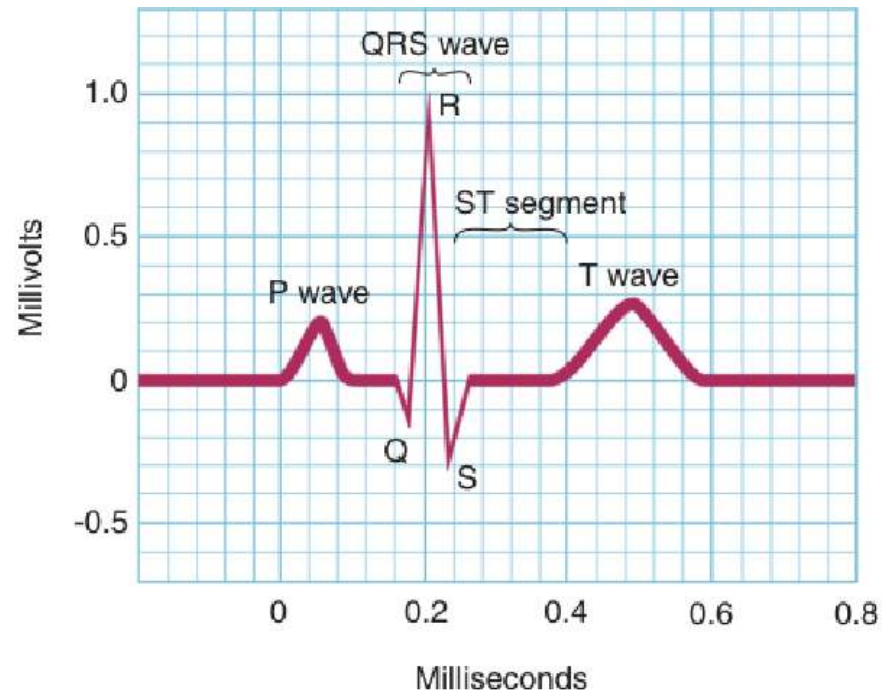
Conduction System of the Heart

(slide 2 of 2)

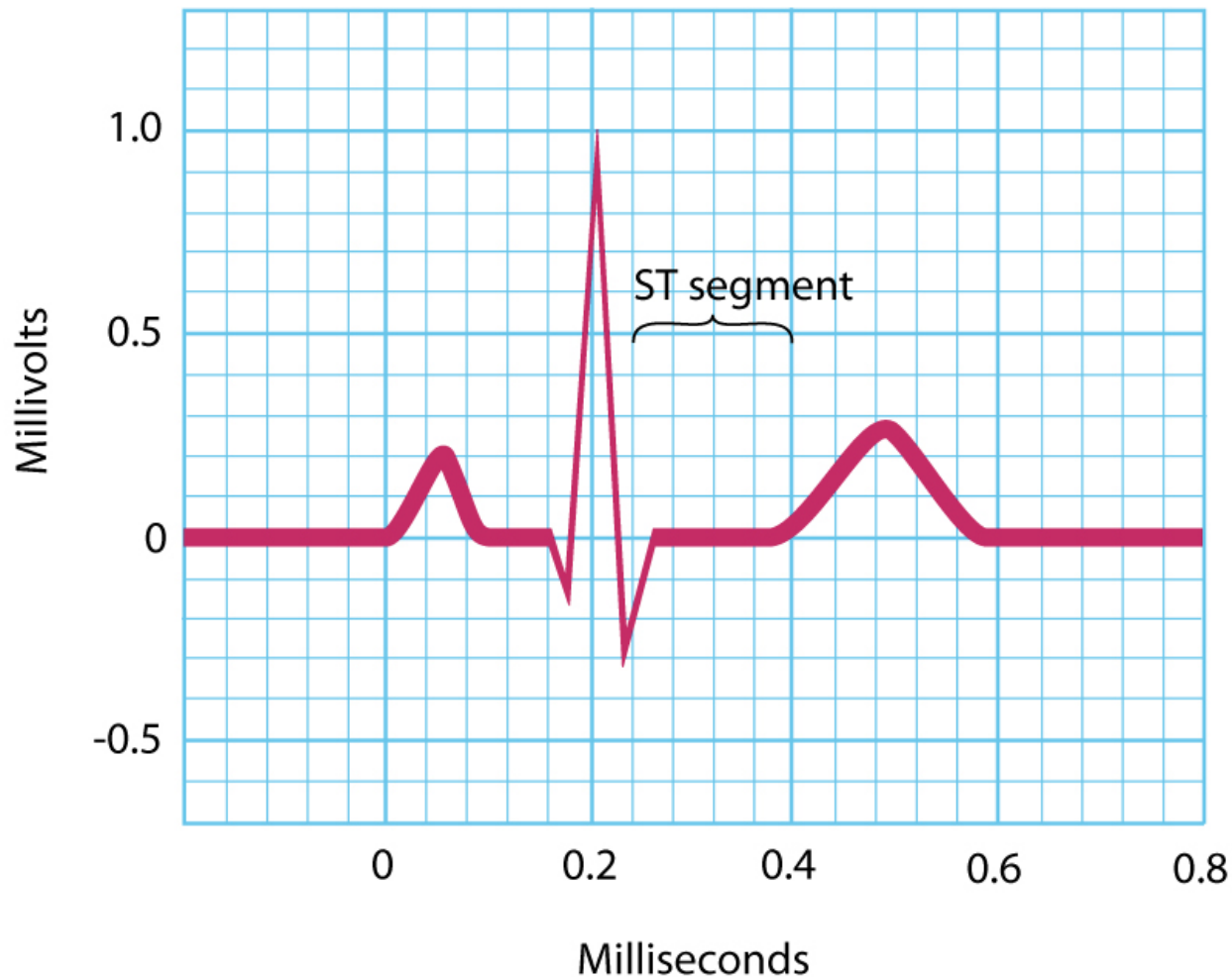
- **Atrioventricular** node (AV node): This sends the excitation wave to a bundle of specialized fibers called the atrioventricular bundle or Bundle of His.
- **Bundle of His (pronounced “hiss”)**: Helps form conduction myofibers that extend to ventricle walls and stimulate them to contract, beginning systole. A short rest period follows.
- The pacemaker begins wave of excitation again.
- ECG or EKG (electrocardiogram): The record used to detect electrical changes in heart muscle as the heart beats.

Electrocardiogram (slide 1 of 3)

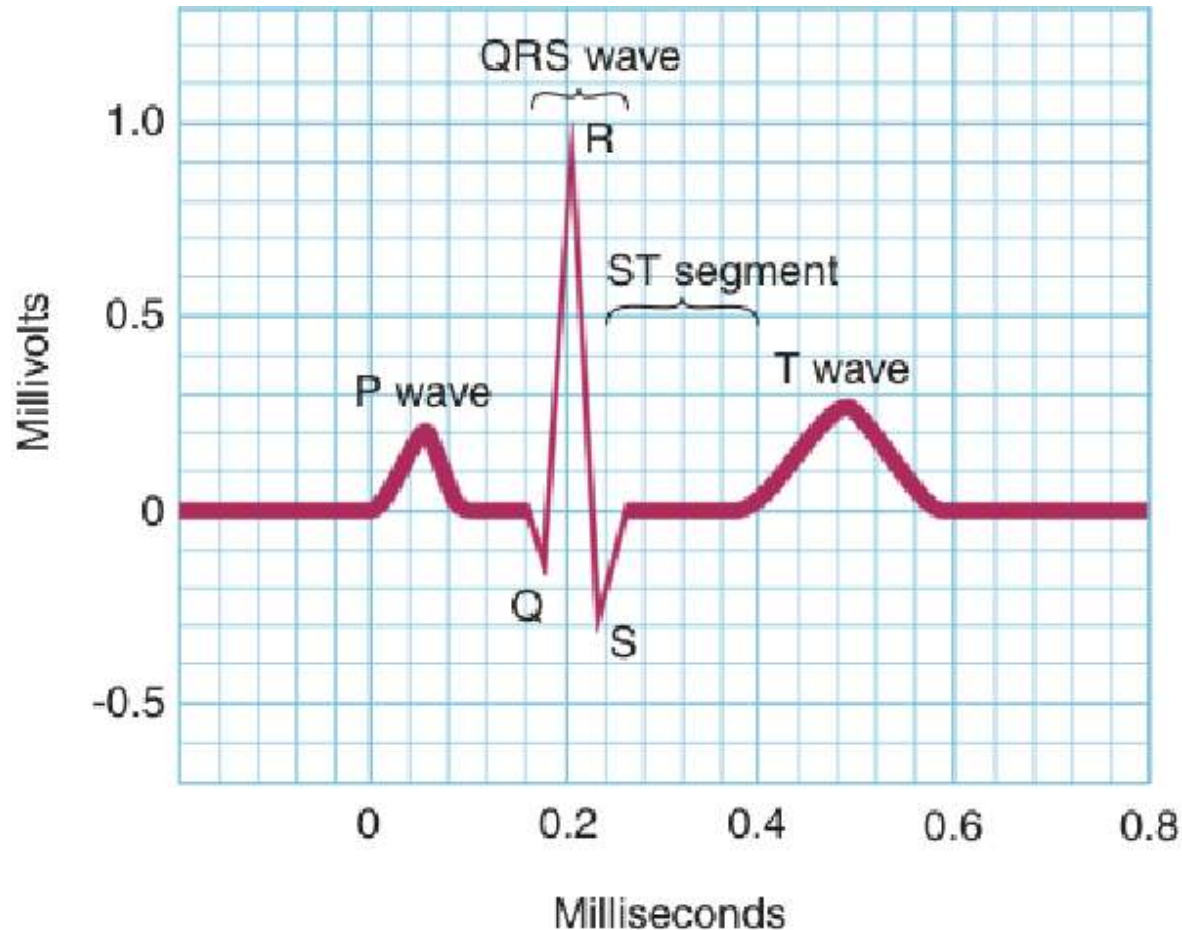
- **P wave** = spread of excitation wave over the atria just before contraction
- **QRS wave** = spread of excitation wave over the ventricles as the ventricles contract;
- **T wave** = electrical recovery and relaxation of ventricles. A heart attack (myocardial infarction or MI) can be recognized by an elevation in the S-T segment of the ECG. Thus, one type of MI is an S-T elevation MI or STEMI



Electrocardiogram (slide 2 of 3)

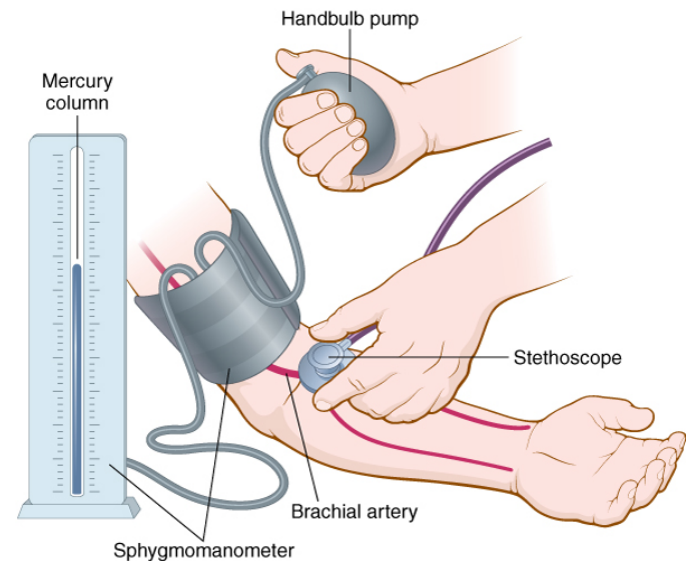


Electrocardiogram (slide 3 of 3)



Blood Pressure

- **Blood pressure:** The force that blood exerts on arterial walls
- Measured using a **sphygmomanometer**
- Expressed as a fraction:
systolic pressure/diastolic pressure
 - Example: 120/80 mm Hg



Vocabulary (Slide 1 of 22)

Term
aorta
apex of the heart
arteriole
artery

Vocabulary (Slide 2 of 22)

Term	Meaning/Definition
aorta	The largest artery in the body
apex of the heart	The lower tip of the heart
arteriole	A small artery
artery	The largest type of blood vessel; carries blood away from the heart to all parts of the body

Vocabulary (Slide 3 of 22)

Term
atrioventricular bundle (bundle of His)
atrioventricular node (AV node)
atrium (<i>plural</i> : atria)

Vocabulary (Slide 4 of 22)

Term	Meaning/Definition
atrioventricular bundle (bundle of His)	Specialized muscle fibers connecting the atria with the ventricles and transmitting electrical impulses between them
atrioventricular node (AV node)	Specialized tissue in the wall between atria; electrical impulses pass from pacemaker through AV node and bundle of His toward ventricles
atrium (<i>plural: atria</i>)	One of two upper chambers of the heart

Vocabulary (Slide 5 of 22)

Term
capillary
carbon dioxide (CO ₂)
coronary arteries
deoxygenated blood

Vocabulary (Slide 6 of 22)

Term	Meaning/Definition
capillary	The smallest blood vessel
carbon dioxide (CO ₂)	Waste gas released by body cells, transported via veins to the heart, and then to lungs for exhalation
coronary arteries	Blood vessels that branch from the aorta and carry oxygen-rich blood to the heart muscle
deoxygenated blood	Oxygen-poor blood

Vocabulary (Slide 7 of 22)

Term
diastole
electrocardiogram
endocardium
endothelium

Vocabulary (Slide 8 of 22)

Term	Meaning/Definition
diastole	Relaxation phase of the heartbeat
electrocardiogram	Record of the electricity flowing through the heart
endocardium	Inner lining of the heart
endothelium	Innermost lining of blood vessels

Vocabulary (Slide 9 of 22)

Term
mitral valve
murmur
myocardium
normal sinus rhythm

Vocabulary (Slide 10 of 22)

Term	Meaning/Definition
mitral valve	Valve between the left atrium and left ventricle; bicuspid valve
murmur	Abnormal swishing sound caused by improper closure of heart valves
myocardium	Muscular, middle layer of the heart
normal sinus rhythm	Heart rhythm originating in sinoatrial node with resting rate of 60-100 beats/min

Vocabulary (Slide 11 of 22)

Term
oxygen
pacemaker (sinoatrial node)
pericardium
pulmonary artery

Vocabulary (Slide 12 of 22)

Term	Meaning/Definition
oxygen	Gas carried by the blood from the lungs to the heart to be pumped via arteries to all body cells
pacemaker (sinoatrial node)	Specialized nervous tissue in the right atrium that begins the heartbeat
pericardium	Double-layered membrane surrounding the heart
pulmonary artery	Artery carrying oxygen-poor blood from the heart to the lungs

Vocabulary (Slide 13 of 22)

Term
pulmonary circulation
pulmonary valve
pulmonary vein

Vocabulary (Slide 14 of 22)

Term	Meaning/Definition
pulmonary circulation	Flow of blood from the heart to the lungs and back to the heart
pulmonary valve	Valve positioned between the right ventricle and the pulmonary artery
pulmonary vein	Carries oxygenated blood from the lungs to the left atrium of the heart

Vocabulary (Slide 15 of 22)

Term
pulse
septum (<i>plural</i> : septa)
sinoatrial node (SA node)

Vocabulary (Slide 16 of 22)

Term	Meaning/Definition
pulse	The beat of the heart as felt through the walls of the arteries
septum (<i>plural: septa</i>)	A partition or wall dividing a cavity, such as that between the right and left atria and right and left ventricles
sinoatrial node (SA node)	The pacemaker of the heart

Vocabulary (Slide 17 of 22)

Term
sphygmomanometer
systemic circulation
systole
tricuspid valve

Vocabulary (Slide 18 of 22)

Term	Meaning/Definition
sphygmomanometer	An instrument to measure blood pressure
systemic circulation	Flow of blood from body tissue to the heart and back
systole	The contraction phase of the heartbeat
tricuspid valve	Located between the right atrium and right ventricle; has three leaflets or cusps

Vocabulary (Slide 19 of 22)

Term
valve
vein

Vocabulary (Slide 20 of 22)

Term	Meaning/Definition
valve	A structure in veins or in the heart that temporarily closes an opening so blood flows in only one direction
vein	A thin-walled vessel that carries blood from body tissues and lungs back to the heart; contains valves to prevent the backflow of blood

Vocabulary (Slide 21 of 22)

Term
vena cava (<i>plural: venae cavae</i>)
ventricle
venule

Vocabulary (Slide 22 of 22)

Term	Meaning/Definition
vena cava (<i>plural: venae cavae</i>)	The largest vein in the body; superior and inferior venae cavae return blood to the right atrium of heart
ventricle	One of two lower chambers of the heart
venule	A small vein

Combining Forms and Terminology (Slide 1 of 4)

Combining Form	Meaning
angi/o	vessel
aort/o	aorta
arter/o	artery
arteri/o	artery
ather/o	yellowish plaque
atri/o	atrium

Combining Forms and Terminology (Slide 2 of 4)

Combining Form	Meaning
brachi/o	arm
cardi/o	heart
cholesterol/o	cholesterol
coron/o	heart
cyan/o	blue
myx/o	mucus

Combining Forms and Terminology (Slide 3 of 4)

Combining Form	Meaning
ox/o	oxygen
pericardi/o	pericardium
phleb/o	vein
sphygm/o	pulse
steth/o	chest
thromb/o	clot

Combining Forms and Terminology (Slide 4 of 4)

Combining Form	Meaning
valvul/o	valve
valv/o	valve
vas/o	vessel
vascul/o	vessel
ven/o, ven/i	vein
ventricul/o	ventricle

QUICK QUIZ (Slide 1 of 4)

1. The double-layered membrane surrounding the heart is the:

- A. pericardium
- B. arteriole
- C. endocardium
- D. endothelium

QUICK QUIZ (Slide 2 of 4)

2. The contraction phase of the heartbeat is called:

- A. diastole
- B. vena cava
- C. systole
- D. septum

Chapter 11

Cardiovascular System

Lesson 11.2

Pathology: Heart (Slide 1 of 7)

Heart

➤ Arrhythmias

- Heart block (atrioventricular block)
- Flutter
- Fibrillation

Pathology: Heart (Slide 2 of 7)

Heart

➤ Congenital heart disease

- Coarctation of the aorta (CoA)
- Patent ductus arteriosus (PDA)
- Septal defects (ASD and VSD)
- Tetralogy of Fallot

Pathology:Heart (Slide 3 of 7)

Heart

- **Congestive heart disease (CHF):** The heart is unable to pump the required amount of blood.
 - In the U.S., primarily the result of high blood pressure and coronary artery disease (see next slide)
 - Results in pulmonary edema
 - Fatal if untreated

Pathology: Heart (Slide 4 of 7)

Heart

➤ **Coronary artery disease (CAD)**

- Atherosclerosis
 - Thrombotic occlusion (occlusive/mural)
 - Ischemia
 - Necrosis
 - Infarction
- Acute Coronary Syndromes (ACS)
 - Unstable angina
 - Myocardial infarction (MI)

Pathology: Heart (Slide 5 of 7)

Heart

➤ **Coronary artery disease (CAD)**

- Drug therapies for CAD

Nitrates (nitroglycerin)

Aspirin

Beta-blockers

ACE inhibitors

Calcium channel blockers

Statins

Pathology: Heart (Slide 6 of 7)

Heart

➤ **Coronary artery disease (CAD)**

- Surgical therapies for CAD

Coronary artery bypass grafting (CABG)

Percutaneous coronary intervention (PCI)

Pathology: Heart (Slide 7 of 7)

Heart

➤ Other pathologic conditions

- Endocarditis (vegetations)
- Hypertensive heart disease
- Mitral valve prolapse (MVP)
- Murmur
- Pericarditis
- Rheumatic heart disease

Pathology:

Blood Vessels

Blood vessels

- Aneurysm
- Deep vein thrombosis (DVT)
- Hypertension (HTN)
- Peripheral vascular disease (PVD)
- Raynaud's disease
- Varicose veins

QUICK QUIZ (Slide 3 of 4)

- 3. Which arrhythmia refers to rapid, random, inefficient, and irregular contractions of the atria and ventricles (350 beats or more per minute)?**
- A. Fibrillation**
 - B. Flutter**
 - C. Bradycardia**

Chapter 11

Lesson 11.3

Laboratory Tests

- BNP test
- Cardia biomarkers
- Lipid test profile
- Lipoprotein electrophoresis

Clinical Procedures: Diagnostic (Slide 1 of 3)

X-ray

- Angiography and arteriography
- Computerized tomography angiography
- Digital subtraction angiography (DSA)
- Electron beam computed tomography (EBCT or EBT)

Ultrasound tests

- Doppler ultrasound
- Echocardiography (ECHO)

Clinical Procedures: Diagnostic (Slide 2 of 3)

Nuclear cardiology

- Positron emission tomography (PET) scan
- Technetium TC 99m scan
- Thallium-201 scan

Magnetic resonance imaging (MRI)

- Cardiac MRI

Clinical Procedures: Diagnostic (Slide 3 of 3)

Other diagnostic procedures

- Cardiac catheterization
- Electrocardiography (ECG, EKG)
- Holter monitoring
- Stress test

Clinical Procedures: Treatment

(Slide 1 of 2)

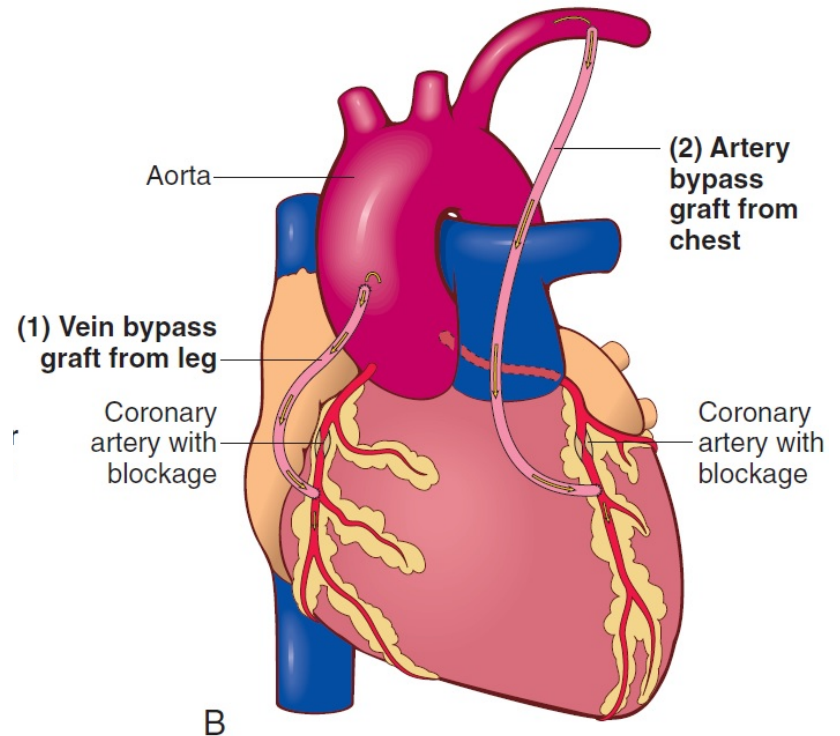
- Cardioversion (defibrillation)
- Endarterectomy
- Extracorporeal circulation
- Heart transplantation
- Thrombolytic therapy
- Transcatheter aortic valve replacement (TAVR)

Clinical Procedures: Treatment (Slide 2 of 2)

Coronary artery bypass graft (CABG) surgery

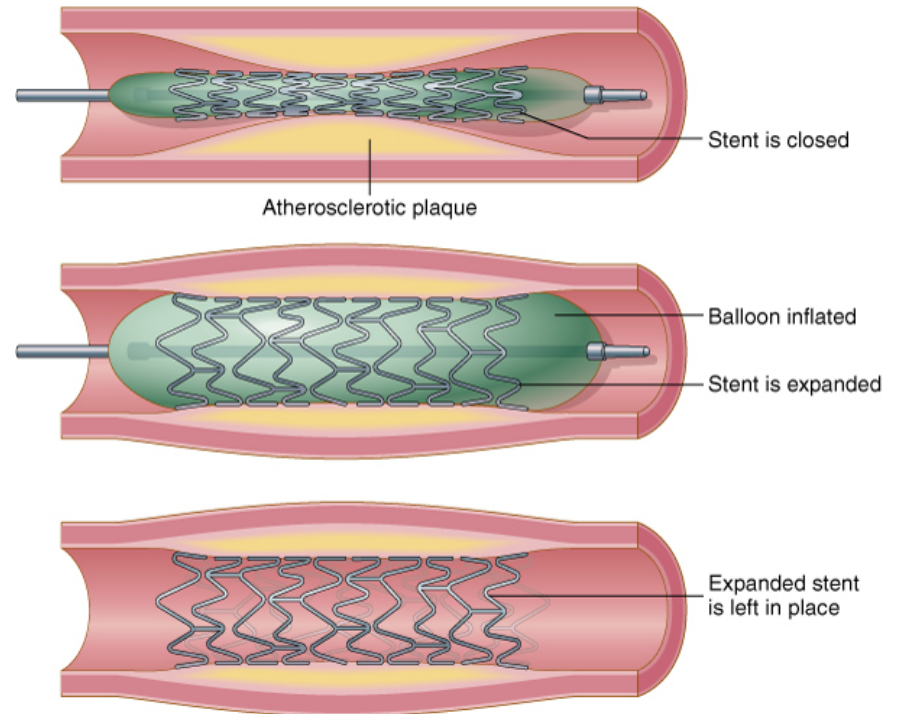
1, A section of a vein is removed from the leg and anastomosed to a coronary artery to bypass an area of arteriosclerotic blockage.

2, An internal mammary artery is grafted to a coronary artery to bypass blockage.



Treatment Procedures

- Percutaneous coronary intervention (PCI)
 - Includes:
 - percutaneous transluminal coronary angioplasty (PTCA),
 - stent placement, laser angioplasty, and
 - atherectomy



Abbreviations (Slide 1 of 9)

Abbreviation	Meaning
AAA	Abdominal aortic aneurysm
ACE inhibitor	Angiotensin-converting enzyme inhibitor
ACLS	Advanced cardiac life support
ACS	Acute coronary syndrome
ADP	Adenosine diphosphate
AED	Automatic external defibrillator
AF, a-fib	Atrial fibrillation
AICD	Automatic implantable cardioverter-defibrillator
AMI	Acute myocardial infarction
ARVD	Arrhythmogenic right ventricular dysplasia
ARB	angiotension II receptor blocker
AS	Aortic stenosis
ASD	Atrial septal defect

Abbreviations (Slide 2 of 9)

Abbreviation	Meaning
AV, A-V	Atrioventricular
AVR	Aortic valve replacement
BBB	Bundle branch block
BNP	Brain natriuretic peptide
BP	Blood pressure
CABG	Coronary artery bypass grafting
CAD	Coronary artery disease
CCU	Coronary care unit
Cath	Catheterization
CHF	Congestive heart failure
CK	Creatinine kinase
CoA	Coarctation of the aorta

Abbreviations (Slide 3 of 9)

Abbreviation	Meaning
CPR	Cardiopulmonary resuscitation
CRT	Cardiac resynchronization therapy
CTNI or cTnI; CTNT or cTnT	Cardiac troponin-I and cardiac troponin-T
DES	Drug-eluting stent
DSA	Digital subtraction angiography
DVT	Deep vein thrombosis
ECMO	Extracorporeal membrane oxygenation
ECG; also EKG	Electrocardiography
ECHO	Echocardiography
EF	Ejection fraction
EPS	Electrophysiology study

Abbreviations (Slide 4 of 9)

Abbreviation	Meaning
ETT	Exercise tolerance test
ETT-MIBI	Exercise tolerance test combined with a radioactive tracer (sestamibi) scan
EVAR	Endovascular aneurysm repair
FFR	Fractional flow reserve
HDL	High-density lipoprotein
hsCRP	High sensitivity C-reactive protein
HTN	Hypertension (high blood pressure)
IABP	Intra-aortic balloon pump

Abbreviations (Slide 5 of 9)

Abbreviation	Meaning
ICD	Implantable cardioverter-defibrillator
IVUS	Intravascular ultrasound
LAD	Left anterior descending (coronary artery)
LDL	Low-density lipoprotein
LMWH	Low-molecular-weight heparin
LV	Left ventricle
LVAD	Left ventricular assist device

Abbreviations (Slide 6 of 9)

Abbreviation	Meaning
LVH	Left ventricular hypertrophy
MI	Myocardial infarction
MUGA	Multiple-gated acquisition scan
MVP	Mitral valve prolapse
NSR	Normal sinus rhythm
NSTEMI	Non-ST elevation myocardial infarction

Abbreviations (Slide 7 of 9)

Abbreviation	Meaning
NT-proBNP	N-terminal pro-peptide of BNP
PAC	Premature atrial contraction
PCI	Percutaneous coronary intervention
PDA	Patent ductus arteriosus; posterior descending artery
PE	Pulmonary embolus
PVC	Premature ventricular contraction
SA, S-A node	Sinoatrial node

Abbreviations (Slide 8 of 9)

Abbreviation	Meaning
SCD	Sudden cardiac death
SOB	Shortness of breath
SPECT	Single photon emission computed tomography
STEMI	ST elevation myocardial infarction
SVT	Supraventricular tachycardia
TAVR	Transcatheter aortic valve replacement
TEE	Transesophageal echocardiography
TEVAR	Thoracic endovascular aneurysm repair

Abbreviations (Slide 9 of 9)

Abbreviation	Meaning
TGA	Transposition of the great arteries
tPA	Tissue-type plasminogen activator
UA	Unstable angina
VF	Ventricular fibrillation
VSD	Ventricular septal defect
VT	Ventricular tachycardia
WPW	Wolff-Parkinson-White syndrome

QUICK QUIZ (Slide 4 of 4)

- 4. Which procedure involves insertion of a balloon-tipped catheter into a coronary artery?**
- A. Thrombolytic therapy
 - B. Coronary artery bypass grafting
 - C. Percutaneous coronary intervention
 - D. Endarterectomy

Review Sheet – Combining Forms

(Slide 1 of 8)

Combining Form
aneurysm/o
angi/o
aort/o
arter/o
arteri/o
ather/o
atri/o

Review Sheet – Combining Forms

(Slide 2 of 8)

Combining Form	Meaning
aneurysm/o	aneurysm
angi/o	vessel
aort/o	aorta
arter/o	artery
arteri/o	artery
ather/o	yellowish plaque
atri/o	atrium

Review Sheet – Combining Forms

(Slide 3 of 8)

Combining Form
axill/o
brachi/o
cardi/o
cholesterol/o
coron/o
cyan/o
isch/o
my/o
myx/o

Review Sheet – Combining Forms

(Slide 4 of 8)

Combining Form	Meaning
axill/o	armpit
brachi/o	arm
cardi/o	heart
cholesterol/o	cholesterol
coron/o	heart
cyan/o	blue
isch/o	to hold back
my/o	muscle
myx/o	mucus

Review Sheet – Combining Forms

(Slide 5 of 8)

Combining Form
ox/o
pericardi/o
phleb/o
pulmon/o
rrhythm/o
sphygm/o
steth/o
thromb/o

Review Sheet – Combining Forms

(Slide 6 of 8)

Combining Form	Meaning
ox/o	oxygen
pericardi/o	pericardium
phleb/o	vein
pulmon/o	lung
rrhythm/o	rhythm
sphygm/o	pulse
steth/o	chest
thromb/o	clot

Review Sheet – Combining Forms

(Slide 7 of 8)

Combining Form
valvul/o
valv/o
vas/o
vascul/o
ven/o, ven/i
ventricul/o

Review Sheet – Combining Forms

(Slide 8 of 8)

Combining Form	Meaning
valvul/o	valve
valv/o	valve
vas/o	vessel
vascul/o	vessel
ven/o, ven/i	vein
ventricul/o	ventricle

Review Sheet – Suffixes (Slide 1 of 4)

Suffix
-constriction
-dilation
-emia
-graphy
-lysis
-megaly

Review Sheet – Suffixes (Slide 2 of 4)

Suffix	Meaning
-constriction	narrowing
-dilation	widening; stretching; expanding
-emia	blood condition
-graphy	process of recording
-lysis	breakdown; separation; destruction; loosening
-megaly	enlargement

Review Sheet – Suffixes (Slide 3 of 4)

Suffix
-meter
-oma
-osis
-plasty
-sclerosis
-stenosis
-tomy

Review Sheet – Suffixes (Slide 4 of 4)

Suffix	Meaning
-meter	measure
-oma	tumor; mass; fluid collection
-osis	condition; usually abnormal
-plasty	surgical repair
-sclerosis	hardening
-stenosis	tightening; structure
-tomy	process of cutting

Review Sheet – Prefixes (Slide 1 of 4)

Prefix
a-, an
brady-
de-
dys-
endo-
hyper-

Review Sheet – Prefixes (Slide 2 of 4)

Prefix	Meaning
a-, an	no; not; without
brady-	slow
de-	lack of; down; less; removal of
dys-	bad; painful; difficult; abnormal
endo-	in; within
hyper-	above; excessive

Review Sheet – Prefixes (Slide 3 of 4)

Prefix
hypo-
inter-
peri-
tachy-
tetra-
tri-

Review Sheet – Prefixes (Slide 4 of 4)

Prefix	Meaning
hypo-	deficient; below; under; less than normal
inter-	between
peri-	surrounding
tachy-	fast
tetra-	four
tri-	three